

**Pro WS
W680M-ACE
SE**



Motherboard

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Contents

Safety information.....	iv
About this guide.....	v
Pro WS W680M-ACE SE specifications summary.....	viii
Package contents.....	xii

Chapter 1: Product Introduction

1.1	Before you proceed	1-1
1.2	Motherboard layout.....	1-2
1.3	Motherboard rear and audio connections	1-18
1.3.1	Rear I/O connection	1-18
1.3.2	Audio I/O connections	1-20

Chapter 2: Basic Setup

2.1	CPU installation.....	2-1
2.2	DIMM installation.....	2-3
2.3	M.2 module installation	2-4
2.4	Motherboard installation	2-9
2.5	BIOS FlashBack™.....	2-10
2.6	Starting up for the first time.....	2-11
2.7	Turning off the computer	2-11

Chapter 3: BIOS and RAID Support

3.1	Knowing UEFI BIOS	3-1
3.2	ASUS EZ Flash Utility	3-2
3.3	ASUS CrashFree BIOS 3.....	3-3
3.4	RAID configurations	3-4

Appendix

Q-Code table.....	A-5
General Notices.....	A-9
Notices for non Wi-Fi model.....	A-12
Warranty.....	A-14
ASUS contact information.....	A-16
Service and Support	A-16

Safety information

Electrical safety

- To prevent electrical shock hazard, disconnect the power cable from the electrical outlet before relocating the system.
- When adding or removing devices to or from the system, ensure that the power cables for the devices are unplugged before the signal cables are connected. If possible, disconnect all power cables from the existing system before you add a device.
- Before connecting or removing signal cables from the motherboard, ensure that all power cables are unplugged.
- Seek professional assistance before using an adapter or extension cord. These devices could interrupt the grounding circuit.
- Ensure that your power supply is set to the correct voltage in your area. If you are not sure about the voltage of the electrical outlet you are using, contact your local power company.
- If the power supply is broken, do not try to fix it by yourself. Contact a qualified service technician or your retailer.

Operation safety

- Before installing the motherboard and adding devices on it, carefully read all the manuals that came with the package.
- Before using the product, ensure all cables are correctly connected and the power cables are not damaged. If you detect any damage, contact your dealer immediately.
- To avoid short circuits, keep paper clips, screws, and staples away from connectors, slots, sockets and circuitry.
- Avoid dust, humidity, and temperature extremes. Do not place the product in any area where it may become wet.
- Place the product on a stable surface.
- If you encounter technical problems with the product, contact a qualified service technician or your retailer.
- Your motherboard should only be used in environments with ambient temperatures between 10°C and 35°C.

Button/Coin Batteries Safety Information



 WARNING		
KEEP OUT OF REACH OF CHILDREN Swallowing can lead to chemical burns, perforation of soft tissue, and death. Severe burns can occur within 2 hours of ingestion. Seek medical attention immediately.		

About this guide

This user guide contains the information you need when installing and configuring the motherboard.

How this guide is organized

This guide contains the following parts:

- **Chapter 1: Product Introduction**
This chapter describes the features of the motherboard and includes descriptions for each part of the motherboard.
- **Chapter 2: Basic Setup**
This chapter lists the basic setup procedures for setting up your motherboard.
- **Chapter 3: BIOS and RAID Support**
This chapter tells how to boot into the BIOS, upgrade BIOS using the EZ Flash Utility and support on RAID.

Where to find more information

Refer to the following sources for additional information and for product and software updates.

1. ASUS website

The ASUS website (www.asus.com) provides updated information on ASUS hardware and software products.

2. Optional documentation

Your product package may include optional documentation, such as warranty flyers, that may have been added by your dealer. These documents are not part of the standard package.

3. MyASUS

MyASUS offers a variety of support features such as helping to troubleshoot issues, optimizing product performance, integrating ASUS software, and recovery drive creation. Please visit <https://www.asus.com/support> for installation guide and FAQ.



MyASUS is only available on selected models, please check your motherboard's specifications summary to see if your motherboard supports MyASUS.

4. Motherboard Installation Guide

Please visit <https://www.asus.com/support> for more information on the Motherboard Installation Guide.



5. Driver and Utilities FAQ

Please visit <https://www.asus.com/support> for more information on downloading and installing drivers and utilities for your motherboard.



6. RAID Configuration Guide

Please visit <https://www.asus.com/support> for more information on the RAID Configuration Guide.



7. BIOS FlashBack™ Feature

Please visit <https://www.asus.com/support> for more information on the BIOS FlashBack™ Feature.



Conventions used in this guide

To ensure that you perform certain tasks properly, take note of the following symbols used throughout this user guide.



CAUTION: Information to prevent damage to the components and injuries to yourself when trying to complete a task.



IMPORTANT: Instructions that you **MUST** follow to complete a task.



NOTE: Tips and additional information to help you complete a task.

Pro WS W680M-ACE SE specifications summary

CPU	Intel® Socket LGA1700 for 13th Gen Intel® Core™ & 12th Gen Intel® Core™ Processors* Supports Intel® Turbo Boost Technology 2.0 and Intel® Turbo Boost Max Technology 3.0** * Refer to www.asus.com for CPU support list. ** Intel® Turbo Boost Max Technology 3.0 support depends on the CPU types.
Chipset	Intel® W680 Chipset
Memory	4 x DIMM, Max. 192GB, DDR5 ECC and Non-ECC Un-buffered Memory* Dual Channel Memory Architecture Supports Intel® Extreme Memory Profile (XMP) OptiMem II * Supported memory types, data rate (speed), and number of DRAM modules vary depending on the CPU and memory configuration, for more information please refer to CPU/Memory Support list under the Support tab of product information site or visit https://www.asus.com/support/. * Non-ECC, un-buffered DDR5 memory supports On-Die ECC function.
Graphics	1 x DisplayPort** 1 x HDMI™ port*** 1 x VGA port from AST2600 * Graphics specifications may vary between CPU types. Please refer to www.intel.com for any updates. ** Supports max. 8K@60Hz as specified in DisplayPort 1.4. *** Supports 4K@60Hz as specified in HDMI 2.1.
Expansion Slots	Intel® 13th & 12th Gen Processors* - 1 x PCIe 5.0 x16 slot Intel® W680 Chipset - 1 x PCIe 4.0 x4 slot - 1 x PCIe 3.0 x1 slot
Storage	Total supports 2 x M.2 slots and 8 x SATA 6Gb/s ports* Intel® 13th & 12th Gen Processors - M.2_1 slot (Key M), type 2242/2260/2280 (supports PCIe 4.0 x4 mode) Intel® W680 Chipset** - M.2_2 slot (Key M), type 2242/2260/2280 (supports PCIe 4.0 x4 mode) - 4 x SATA 6Gb/s ports - 1 x SlimSAS Slot Support SlimSAS NVMe device (supports PCIe 4.0 x4 mode and up to 4 SATA devices)*** * Intel® Rapid Storage Technology supports PCIe RAID 0/1/5/10, SATA RAID 0/1/5/10. ** Intel® Rapid Storage Technology supports Intel® Optane Memory H Series on PCH attached M.2 slots. *** SlimSAS slot can support up to 4 SATA devices via a transfer cable. The cable is purchased separately.
Ethernet	2 x Intel® 2.5Gb Ethernet 1 x Realtek 1Gb Ethernet dedicated for BMC ASUS LANGuard

(continued on the next page)

Pro WS W680M-ACE SE specifications summary

USB	Rear USB (Total 6 ports) - 2 x USB 3.2 Gen 2 (10G) ports (1 x Type-A, 1 x Type-C®) - 4 x USB 3.2 Gen 1 (5G) ports (4 x Type-A) Front USB (Total 7 ports) - 1 x USB 3.2 Gen 2x2 (20G) connector (supports USB Type-C®) - 1 x USB 3.2 Gen 1 (5G) header supports additional 2 USB 3.2 Gen 1 (5G) ports - 2 x USB 2.0 headers support additional 4 USB 2.0 ports
Audio	Realtek 7.1 Surround Sound High Definition Audio CODEC - Supports: Jack-detection, Multi-streaming, Front Panel Jack-retasking - Supports up to 24-Bit/192 kHz playback Audio Features - Audio Shielding - Premium audio capacitors - Dedicated audio PCB layers * A chassis with an HD audio module in the front panel is required to support 7.1 Surround Sound audio output.
Back Panel I/O Ports	2 x USB 3.2 Gen 2 (10G) ports (1 x Type-A, 1 x Type-C®) 4 x USB 3.2 Gen 1 (5G) ports (4 x Type-A) 1 x DisplayPort 1 x HDMI™ port 1 x VGA port from AST2600 2 x Intel® 2.5Gb Ethernet ports 1 x Realtek 1Gb Ethernet dedicated for BMC 3 x Audio jacks 1 x BIOS Flashback™ button
Internal I/O connectors	Fan and Cooling related 1 x 4-pin CPU Fan header 1 x 4-pin CPU OPT Fan header 1 x 4-pin W_PUMP+ header 3 x 4-pin Chassis Fan headers Power related 1 x 24-pin Main Power connector 1 x 8-pin +12V Power connector Storage related 2 x M.2 slots (Key M) 4 x SATA 6Gb/s ports 1 x SlimSAS port USB 1 x USB 3.2 Gen 2x2 (20G) connector (supports USB Type-C®) 1 x USB 3.2 Gen 1 (5G) header supports 2 additional USB 3.2 Gen 1 (5G) ports 2 x USB 2.0 headers support 4 additional USB 2.0 ports

(continued on the next page)

Pro WS W680M-ACE SE specifications summary

Internal I/O connectors	Miscellaneous 1 x T-sensor header 1 x Clear CMOS header 1 x COM Port header 1 x Front Panel Audio header (AAFP) 1 x TPM header (14-1pin) 1 x 20-3 pin System Panel header with Chassis intrude function
BMC Related	1 x BMC LED and switch 1 x VGA switch 1 x BMC LAN fixed IP switch 1 x IPMI switch 1 x Q_PSU switch 1 x PSU SMBus header 1 x VPP_I2C header 1 x microSD card socket 1 x BMC_T_SENSOR header 1 x Location button header 1 x Location LED and header 1 x Message LED and header
Special Features	ASUS 5X PROTECTION III - DIGI+ VRM (- Digital power design with DrMOS) - ESD Guards - LANGuard - Overvoltage Protection - SafeSlot - Stainless-Steel Back I/O ASUS Q-Design - Q-Code - M.2 Q-Latch - Q-Connector - Q-DIMM - Q-LED (CPU [red], DRAM [yellow], VGA [white], Boot Device [yellow green]) - Q-Slot ASUS Thermal Solution - M.2 heatsink - VRM heatsink design

(continued on the next page)

Pro WS W680M-ACE SE specifications summary

Special Features	ASUS EZ DIY - CPU Socket lever protector - SafeDIMM - SafeSlot - BIOS FlashBack™ button - BIOS FlashBack™ LED Bespoke Motherboard Design & Business Focused Features - 24/7 Reliability
Software Features	ASUS Exclusive Software Armoury Crate - Fan Xpert 4 - Power Saving ASUS CPU-Z Adobe Creative Cloud (Free Trial) Norton 360 Deluxe (60 Days Free Trial) WinRAR IT Management software supported - ASUS Control Center Express(ACCE) Intel vPro UEFI BIOS ASUS EZ DIY - ASUS CrashFree BIOS 3 - ASUSTek. EzFlash Utility
BIOS	256 Mb Flash ROM, UEFI AMI BIOS
BIOS CAP Filename	Pro WS W680M-ACE SE: PWW680MA.CAP
Manageability	WOL by PME, PXE
Operating System	Windows® 11
	Windows® 10 64-bit
Form Factor	Micro-ATX Form Factor
	9.6 in x 9.6 in (24.4 cm x 24.4 cm)



Specifications are subject to change without notice. Please refer to the ASUS website for the latest specifications.

Package contents

Check your motherboard package for the following items.

Motherboard	1 x Pro WS W680M-ACE SE motherboard
Cables	2 x SATA 6Gb/s cables
Miscellaneous	1 x Q-connector
	1 x M.2 Rubber Packages
	2 x Screw package for M.2 SSD
	1 x I/O Shield
Documentation	1 x User guide
	1 x ACC Express Activation Key Card



- If any of the above items is damaged or missing, contact your retailer.
- Items not listed in the Package contents list above are purchased separately and do not come bundled with your motherboard package.

Product Introduction

1

1.1 Before you proceed

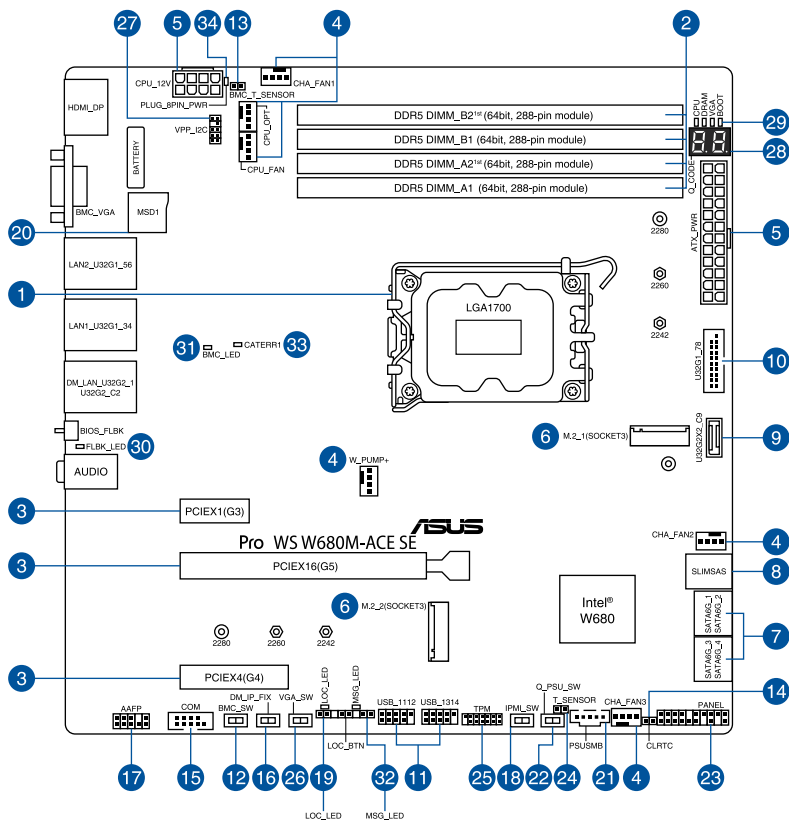
Take note of the following precautions before you install motherboard components or change any motherboard settings.



-
- Unplug the power cord from the wall socket before touching any component.
 - Before handling components, use a grounded wrist strap or touch a safely grounded object or a metal object, such as the power supply case, to avoid damaging them due to static electricity.
 - Hold components by the edges to avoid touching the ICs on them.
 - Whenever you uninstall any component, place it on a grounded antistatic pad or in the bag that came with the component.
 - Before you install or remove any component, ensure that the power supply is switched off or the power cord is detached from the power supply. Failure to do so may cause severe damage to the motherboard, peripherals, or components.
-



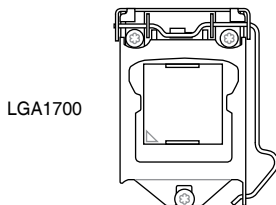
-
- The pin definitions in this chapter are for reference only. The pin names depend on the location of the header/jumper/connector.
 - The illustrations for this chapter are for reference only. The WiFi module is only available on selected models.
-



Layout contents	Page
1. CPU socket	1-4
2. DIMM slots	1-4
3. Expansion slots	1-6
4. Fan and Pump headers	1-7
5. Power connectors	1-8
6. M.2 slot	1-8
7. SATA 6Gb/s port	1-9
8. SlimSAS port	1-9
9. USB 3.2 Gen 2x2 (20G) Type-C® Front Panel connector	1-9
10. USB 3.2 Gen 1 (5G) header	1-10
11. USB 2.0 header	1-10
12. BMC switch	1-10
13. BMC Thermal Sensor header	1-10
14. Clear CMOS header	1-11
15. COM Port header	1-11
16. Fixed Dedicated BMC LAN IP switch	1-11
17. Front Panel Audio header	1-12
18. IPMI switch	1-12
19. Location button and LED headers	1-12
20. microSD Card slot	1-12
21. Power Supply SMBus connector	1-13
22. SMART PSU switch	1-13
23. System Panel header	1-14
24. Thermal Sensor header	1-15
25. TPM header	1-15
26. VGA Switch	1-15
27. VPP_I2C header	1-16
28. Q-Code LED	1-16
29. Q LEDs	1-16
30. BIOS FlashBack™ LED	1-16
31. BMC LED	1-17
32. Message LED	1-17
33. Processor Catastrophic Error LED	1-17
34. 8-pin Power Plug LED	1-17

1. CPU socket

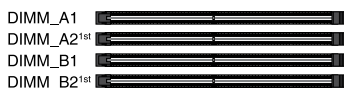
The motherboard comes with a LGA1700 socket designed for 13th Gen Intel® Core™ & 12th Gen Intel® Core™ Processors.



- Keep the cap after installing the motherboard. ASUS will process Return Merchandise Authorization (RMA) requests only if the motherboard comes with the cap on the CPU socket.
- The product warranty does not cover damage to the socket contacts resulting from incorrect CPU installation/removal, or misplacement/loss/incorrect removal of the PnP cap.

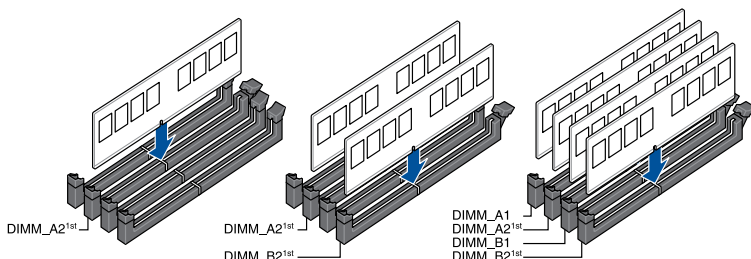
2. DIMM slots

The motherboard comes with Dual Inline Memory Modules (DIMM) slots designed for DDR5 (Double Data Rate 5) memory modules.



A DDR5 memory module is notched differently from a DDR, DDR2, DDR3, or DDR4 module. DO NOT install a DDR, DDR2, DDR3, or DDR4 memory module to the DDR5 slot.

Recommended memory configurations



Memory configurations

You may install ECC and Non-ECC DDR5 DIMMs into the DIMM sockets.



- You may install varying memory sizes in the DIMM channels. The system maps the total size of the lower-sized channel for the dual-channel configuration. Any excess memory from the higher-sized channel is then mapped for single-channel operation.
- The default memory operation frequency is dependent on its Serial Presence Detect (SPD), which is the standard way of accessing information from a memory module. Under the default state, some memory modules for overclocking may operate at a lower frequency than the vendor-marked value.
- For system stability, use a more efficient memory cooling system to support a full memory load or overclocking condition.
- Always install the DIMMs with the same CAS Latency. For an optimum compatibility, we recommend that you install memory modules of the same version or data code (D/C) from the same vendor. Check with the vendor to get the correct memory modules.
- Visit the ASUS website for the latest QVL.

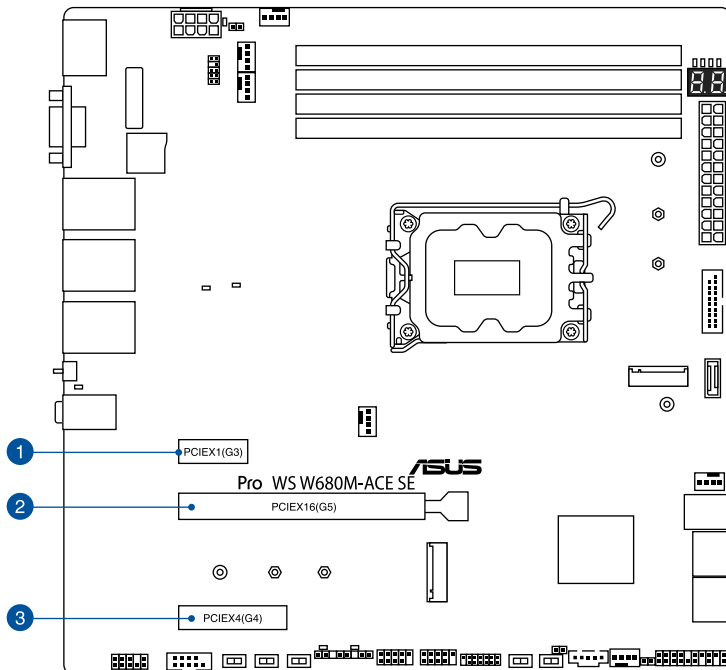
3. Expansion slots



Unplug the power cord before adding or removing expansion cards. Failure to do so may cause you physical injury and damage motherboard components.

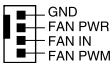


To install a PCIe expansion card, please refer to the **Motherboard Installation Guide** on the ASUS support site.



4. Fan and Pump headers

The Fan and Pump headers allow you to connect fans or pumps to cool the system.

CPU_FAN	CHA_FAN2	
CPU_OPT	CHA_FAN3	
CHA_FAN1	W_PUMP+	



- DO NOT forget to connect the fan cables to the fan headers. Insufficient air flow inside the system may damage the motherboard components. These are not jumpers! Do not place jumper caps on the fan headers!
- Ensure the cable is fully inserted into the header.



For water cooling kits, connect the pump connector to the **W_PUMP+** header.

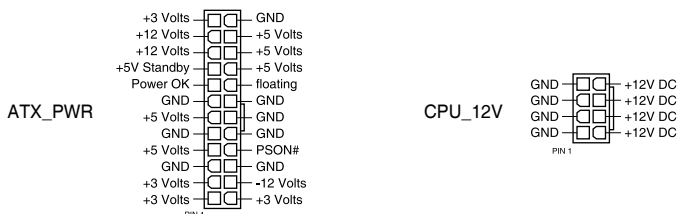


W_PUMP+ function support depends on water cooling device.

Header	Max. Current	Max. Power	Default Speed	Shared Control
CPU_FAN	3A	36W	Q-Fan Controlled	A
CPU_OPT	3A	36W	Q-Fan Controlled	A
CHA_FAN1	3A	36W	Q-Fan Controlled	-
CHA_FAN2	3A	36W	Q-Fan Controlled	-
CHA_FAN3	3A	36W	Q-Fan Controlled	-
W_PUMP+	3A	36W	Full Speed	-

5. Power connectors

These Power connectors allow you to connect your motherboard to a power supply. The power supply plugs are designed to fit in only one orientation, find the proper orientation and push down firmly until the power supply plugs are fully inserted.



- Ensure to connect the 8-pin power plug.



- For a fully configured system, we recommend that you use a power supply unit (PSU) that complies with ATX 12V Specification 2.0 (or later version) and provides a minimum power of 350 W.
- We recommend that you use a PSU with a higher power output when configuring a system with more power-consuming devices. The system may become unstable or may not boot up if the power is inadequate.

6. M.2 slot

The M.2 slot allows you to install M.2 devices such as M.2 SSD modules.

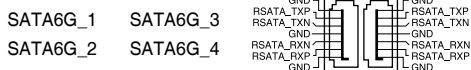
M.2_1(SOCKET3)
M.2_2(SOCKET3)



- **Intel® 13th & 12th Gen Processors:**
 - M.2_1 slot (Key M), type 2242/2260/2280 (supports PCIe 4.0 x4 mode)
- **Intel® W680 Chipset:**
 - M.2_2 slot (Key M), type 2242/2260/2280 (supports PCIe 4.0 x4 mode)
- To enable Intel® Optane™ Memory (Hybrid Storage device), it must be installed in PCH-attached slots with Intel® Rapid Storage Technology.
- Intel® Rapid Storage Technology supports NVMe RAID 0/1/5, SATA RAID 0/1/5/10.

7. SATA 6Gb/s port

The SATA 6Gb/s port allows you to connect SATA devices such as optical disc drives and hard disk drives via a SATA cable.



- If you installed SATA storage devices to the **SATA6G_1-4** ports, you can create a RAID 0, 1, 5, and 10 configuration with the Intel® Rapid Storage Technology through the onboard Intel® W680 chipset.
- To install a SATA device, please refer to the **Motherboard Installation Guide** on the ASUS support site.
- Before creating a RAID set, refer to the **RAID Configuration Guide**. You can download the **RAID Configuration Guide** from the ASUS website.

8. SlimSAS port

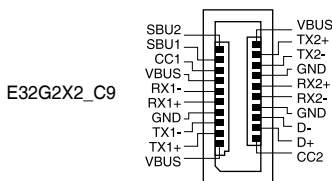
The SlimSAS port allows you to connect PCIe 4.0 x4 NVMe storage devices, and can support up to 4 SATA devices using a transfer cable.



Supports Intel® Virtual RAID on CPU (Intel® VROC) and Intel Volume Management Device (Intel®VMD)

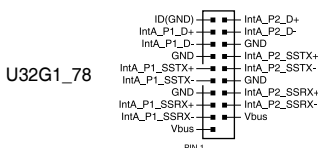
9. USB 3.2 Gen 2x2 (20G) Type-C® Front Panel connector

The USB 3.2 Gen 2x2 (20G) Type-C® connector allows you to connect a USB 3.2 Gen 2x2 (20G) Type-C® module for additional USB 3.2 Gen 2x2 (20G) ports on the front panel. The USB 3.2 Gen 2x2 (20G) Type-C® connector provides data transfer speeds of up to 20 Gb/s.



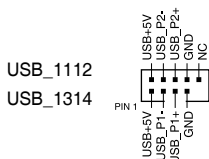
10. USB 3.2 Gen 1 (5G) header

The USB 3.2 Gen 1 (5G) header allows you to connect a USB 3.2 Gen 1 (5G) module for additional USB 3.2 Gen 1 (5G) ports. The USB 3.2 Gen 1 (5G) header provides data transfer speeds of up to 5 Gb/s.



11. USB 2.0 header

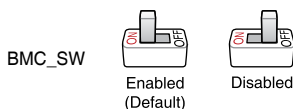
The USB 2.0 header allows you to connect a USB module for additional USB 2.0 ports. The USB 2.0 header provides data transfer speeds of up to 480 Mb/s.



DO NOT connect a 1394 cable to the USB connectors. Doing so will damage the motherboard!

12. BMC switch

The BMC switch allows you to enable or disable the BMC.



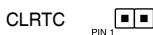
13. BMC Thermal Sensor header

The BMC Thermal Sensor header allows you to connect a sensor to monitor the temperature of the devices and the critical components inside the motherboard through BMC. Connecting the T sensor cables and setting **BMC_SW** to enabled will allow you to view the sensor readings in both the BIOS and on the web UI.



14. Clear CMOS header

The Clear CMOS header allows you to clear the Real Time Clock (RTC) RAM in the CMOS, which contains the date, time, system passwords, and system setup parameters.



To erase the RTC RAM:

1. Turn OFF the computer and unplug the power cord.
2. Short-circuit pin 1-2 with a metal object or jumper cap for about 5-10 seconds.
3. Plug the power cord and turn ON the computer.
4. Hold down the key during the boot process and enter BIOS setup to re-enter data.



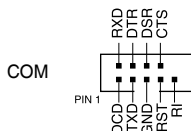
DO NOT short-circuit the pins except when clearing the RTC RAM. Short-circuiting or placing a jumper cap will cause system boot failure!



If the steps above do not help, remove the onboard button cell battery and short the two pins again to clear the CMOS RTC RAM data. After clearing the CMOS, reinstall the button cell battery.

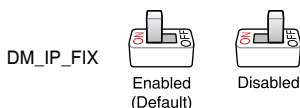
15. COM Port connector

The COM (Serial) Port connector allows you to connect a COM port module. Connect the COM port module cable to this connector, then install the module to a slot opening on the system chassis.



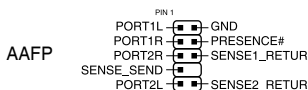
16. Fixed Dedicated BMC LAN IP switch

The Fixed Dedicated BMC LAN IP switch allows you to set a fixed IP (10.10.10.10) when set to enabled.



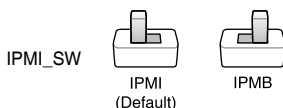
17. Front Panel Audio header

The Front Panel Audio header is for a chassis-mounted front panel audio I/O module that supports HD Audio. Connect one end of the front panel audio I/O module cable to this header.



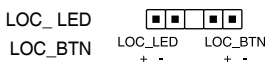
18. IPMI switch

The IPMI switch allows you to switch I2C BUS for instances where I2C may clash due to all PCIe slots being occupied with the same expansion cards.



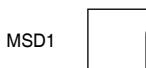
19. Location button and LED headers

The Location button and LED headers allow you to connect a locator button and locator LED on the front panel. This button queries the state of the system locator, and the LEDs will light up when the Locator button is pressed.



20. microSD Card slot

The microSD Card slot allows you to install a microSD memory card v2.00 (SDHC) / v3.00 (SDXC) to log BMC events.



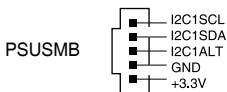
- Disconnect all power (including redundant PSUs) from the existing system before you add or remove a memory card, then reboot the system to access the memory card.
- Some memory cards may not be compatible with your motherboard. Ensure that you use only compatible memory cards to prevent loss of data, damage to your device, or memory card, or both.



The MicroSD Slot is only supported with BMC Function and not supported for normal use under the OS.

21. Power Supply SMBus connector

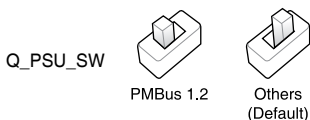
The Power Supply SMBus connector allows you to connect the SMBus (System Management Bus) to the PSU (power supply unit) to read the PSU information. Devices communicate with an SMBus host and/or other SMBus devices using the SMBus interface.



Power supply is required to meet PMBus specification and customized BMC FW may be needed. Please contact ASUS if you need further support

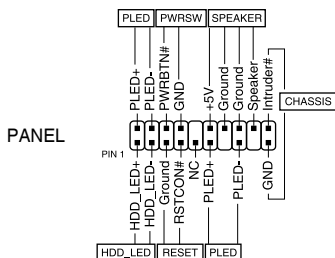
22. SMART PSU switch

This switch allows you to select PSU PMBus version.



23. System Panel header

The System Panel header supports several chassis-mounted functions.



- **System Power LED header (PLED)**

The 2-pin and/or 3-1 pin headers allow you to connect the System Power LED. The System Power LED lights up when the system is connected to a power source, or when you turn on the system power, and blinks when the system is in sleep mode.

- **Storage Device Activity LED header (HDD_LED)**

The 2-pin header allows you to connect the Storage Device Activity LED. The Storage Device Activity LED lights up or blinks when data is read from or written to the storage device or storage device add-on card.

- **System Warning Speaker header (SPEAKER)**

The 4-pin header allows you to connect the chassis-mounted system warning speaker. The speaker allows you to hear system beeps and warnings.

- **Power Button/Soft-off Button header (PWRSW)**

The 3-1 pin header allows you to connect the system power button. Press the power button to power up the system, or put the system into sleep or soft-off mode (depending on the operating system settings).

- **Reset button header (RESET)**

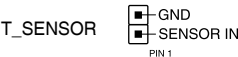
The 2-pin header allows you to connect the chassis-mounted reset button. Press the reset button to reboot the system.

- **Chassis intrusion header (CHASSIS)**

The 2-pin header allows you to connect the chassis-mounted intrusion detection sensor or switch. The chassis intrusion sensor or switch sends a high-level signal to the header when a chassis component is removed or replaced, the signal is then generated as a chassis intrusion event.

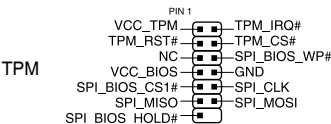
24. Thermal Sensor header

The Thermal Sensor header allows you to connect a sensor to monitor the temperature of the devices and the critical components inside the motherboard. Connect the thermal sensor and place it on the device or the motherboard's component to detect its temperature.



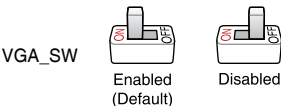
25. TPM header

The TPM header allows you to connect a TPM module, which securely stores keys, digital certificates, passwords, and data. A TPM system also helps enhance network security, protect digital identities, and ensures platform integrity.



26. VGA switch

The VGA switch allows you to enable or disable the onboard VGA controller.



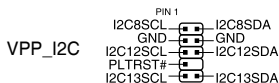
Ensure the ATX power supply is switched off or the power cord is detached from the power supply when enabling or disabling the VGA controller settings using this switch.



- If a VGA Card is installed into a PCI Express x16 Slot, the onboard VGA function will still be enabled.
- BMC Remote Management Function will still be available when VGA controller settings is set to disabled, but the display will be disabled on the client device.

27. VPP_I2C header

The VPP_I2C header is used for the storage backplane with sensor readings.



28. Q-Code LED

The Q-Code LED design provides you with a 2-digit error code that displays the system status.



- The Q-Code LEDs provide the most probable cause of an error code as a starting point for troubleshooting. The actual cause may vary from case to case.
- Please refer to the Q-Code table in the **Appendix** section for more details.

29. Q-LEDs

The Q-LEDs check key components (CPU, DRAM, VGA, and booting devices) during the motherboard booting process. If an error is found, the critical component's LED stays lit up until the problem is solved.

CPU (RED)	☐
DRAM(YELLOW)	☐
VGA (WHITE)	☐
BOOT (YELLOW GREEN)	☐



The Q-LEDs provide the most probable cause of an error code as a starting point for troubleshooting. The actual cause may vary from case to case.

30. BIOS FlashBack™ LED

The BIOS FlashBack™ LED lights up or blinks to indicate the status of the BIOS FlashBack™.

FLBK_LED ☐



Refer to the **BIOS FlashBack™** section for more information on using the BIOS FlashBack™ feature.

31. BMC LED

The BMC LED works with the ASUS ASMB management device and indicates its initiation status. When the PSU is plugged and the system is OFF, ASUS ASMB management device starts system initiation for about one (1) minute. The BMC LED blinks after system initiation finishes.

BMC_LED 

32. Message LED and header

The 2-pin Message LED header is for the message LED cable that connects to the front message LED. The message LED is controlled by the BMC to indicate an abnormal event occurrence.

MSG_LED 

33. Processor Catastrophic Error LED

The Processor Catastrophic Error LED indicates that the system has experienced a fatal or catastrophic error and cannot continue to operate.

CATERR1 

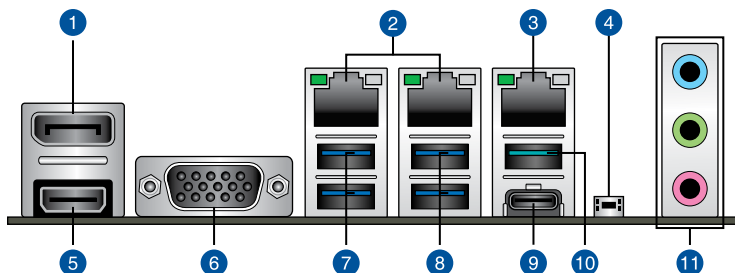
34. 8-pin Power Plug LED

The 8-pin Power Plug LED lights up to indicate that the 8-pin power plug is not connected.

PLUG_8PIN_PWR 

1.3 Motherboard rear and audio connections

1.3.1 Rear I/O connection



Rear panel connectors

1.	DisplayPort
2.	Intel® 2.5Gb Ethernet ports*
3.	Realtek 1Gb Ethernet port dedicated for BMC*
4.	BIOS FlashBack™ button
5.	HDMI™ port
6.	VGA port from AST2600
7.	USB 3.2 Gen 1 (5G) (Blue) Type-A ports 5 and 6
8.	USB 3.2 Gen 1 (5G) (Blue) Type-A ports 3 and 4
9.	USB 3.2 Gen 2 (10G) port Type-C® port C2
10.	USB 3.2 Gen 2 (10G) (Teal blue) port 1
11*.	Audio jacks**

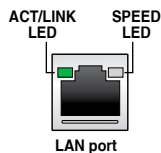
* and **: Refer to the tables on the next page for LAN port LEDs, and audio port definitions.



We strongly recommend that you connect your devices to ports with matching data transfer rate. For example connecting your USB 5Gbps devices to USB 5Gbps ports for faster and better performance for your devices.

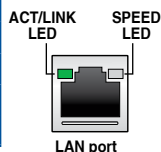
* Intel® 2.5Gb Ethernet port LED indications

Activity Link LED		Speed LED	
Status	Description	Status	Description
OFF	No link	OFF	No link
GREEN	Linked	OFF	100 Mbps / 10 Mbps connection
BLINKING	Data activity	GREEN	2.5 Gbps connection
		ORANGE	1 Gbps connection



* Realtek 1Gb Ethernet port LED indications

Activity Link LED		Speed LED	
Status	Description	Status	Description
OFF	No link	OFF	10 Mbps connection
ORANGE	Linked	ORANGE	100 Mbps connection
BLINKING	Data activity	GREEN	1 Gbps connection



** Audio 2, 4, 5.1 or 7.1-channel configuration

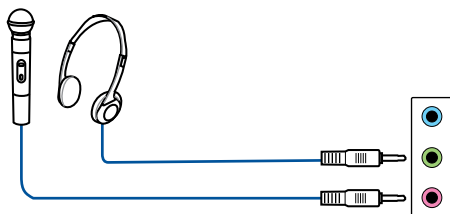
Port	2-channel	4-channel	5.1-channel	7.1-channel
Light Blue (Rear panel)	-	Rear Speaker Out	Rear Speaker Out	Rear Speaker Out
Lime (Rear panel)	Front Speaker Out	Front Speaker Out	Front Speaker Out	Front Speaker Out
Pink/Red (Rear panel)	-		Center/ Subwoofer	Center/ Subwoofer
Lime (Front panel)	-	-	-	Side Speaker Out*
Pink (Front panel)	-	-	-	-

1.3.2 Audio I/O connections

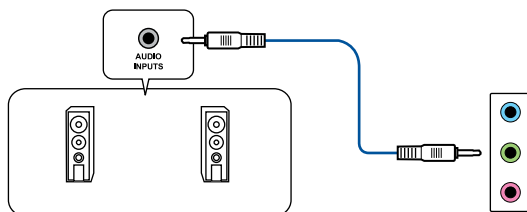
Audio I/O ports



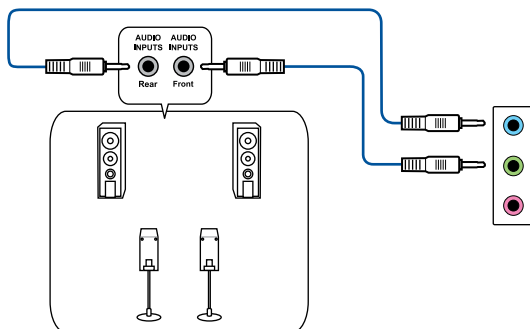
Connect to Headphone and Mic



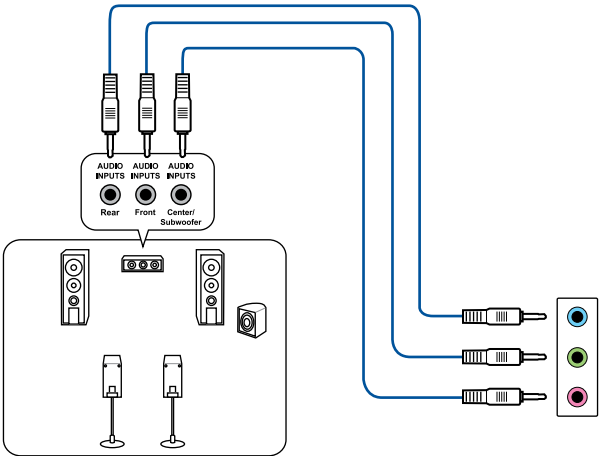
Connect to 2-channel Speakers



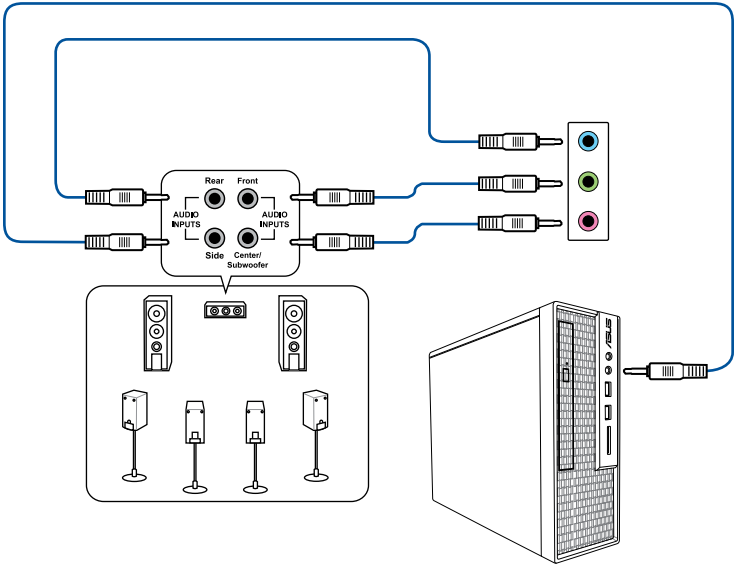
Connect to 4-channel Speakers



Connect to 5.1-channel Speakers



Connect to 7.1-channel Speakers



[illegible]

Basic Setup

2



The installation diagrams in this section are for reference only. The motherboard layout may vary with models, but the installation steps are the same for all models.

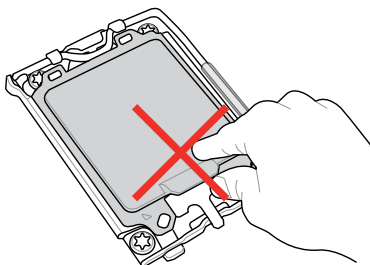
2.1 CPU installation



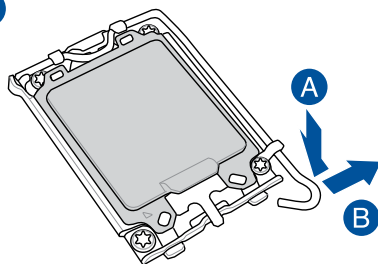
- Ensure that you install the correct CPU designed for LGA1700 socket only. DO NOT install a CPU designed for LGA1155, LGA1156, LGA1151, and LGA1200 sockets on the LGA1700 socket.
- The CPU fits in only one correct orientation. DO NOT force the CPU into the socket to prevent bending the connectors on the socket and damaging the CPU.
- Ensure that all power cables are unplugged before installing the CPU.
- Upon purchase of the motherboard, ensure that the PnP cap is on the socket and the socket contacts are not bent. Contact your retailer immediately if the PnP cap is missing, or if you see any damage to the PnP cap/socket contacts/motherboard components. ASUS will shoulder the cost of repair only if the damage is shipment/transit-related.



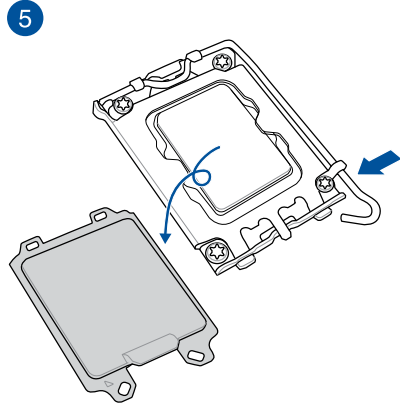
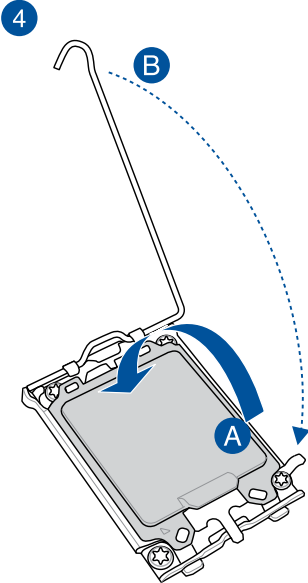
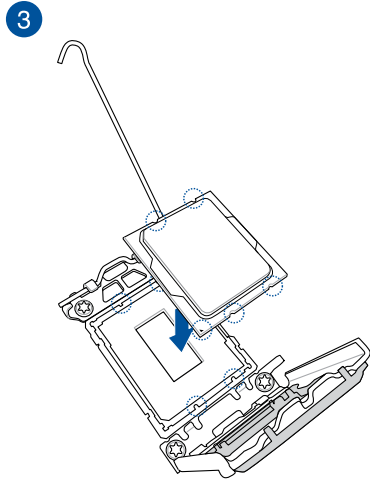
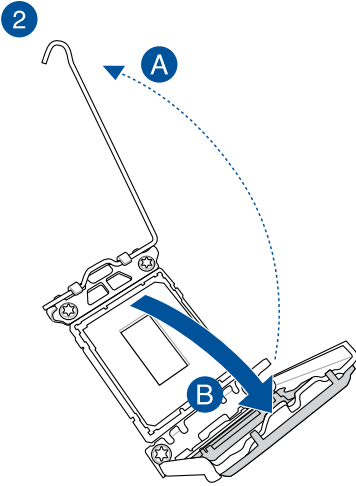
Install a heatsink or AIO cooler after installing the CPU. Please refer to the **Motherboard Installation Guide** on the ASUS support site, or to the user manual of the heatsink/AIO cooler for steps on installing the heatsink/AIO cooler.



1

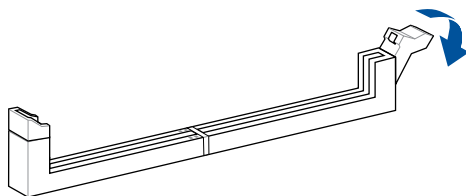


Take caution when lifting the load lever, ensure to hold onto the load lever when releasing the load lever. Letting go of the load lever immediately after releasing it may cause the load lever to spring back and cause damage to your motherboard.

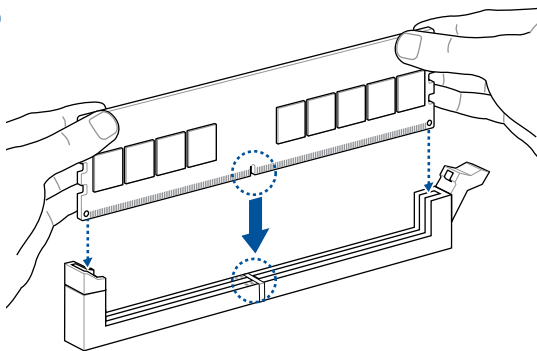


2.2 DIMM installation

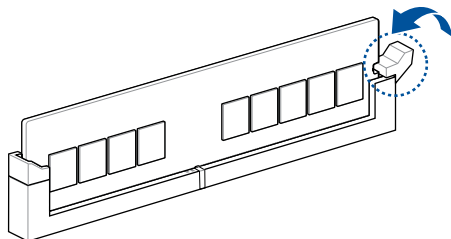
1



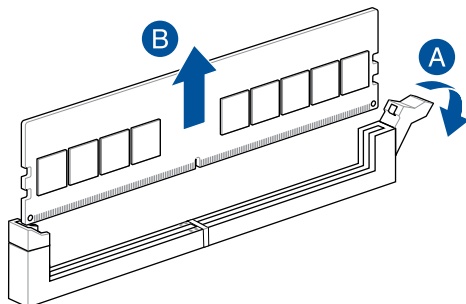
2



3



DIMM removal



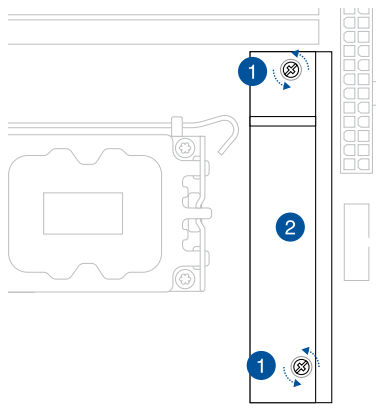
2.3 M.2 module installation



- The illustrations only show the installation steps for a 2242 M.2 slot, the steps are the same for the other M.2 slots.
- Use a Phillips screwdriver when removing or installing the screws or screw stands mentioned in this section.
- If the thermal pad on the M.2 heatsink becomes damaged, we recommend replacing it with the bundled thermal pad or a thermal pad with a thickness of 1.25mm.
- Supported M.2 type varies per motherboard.

For M.2_1 (Type 2242 / 2260 / 2280 M.2)

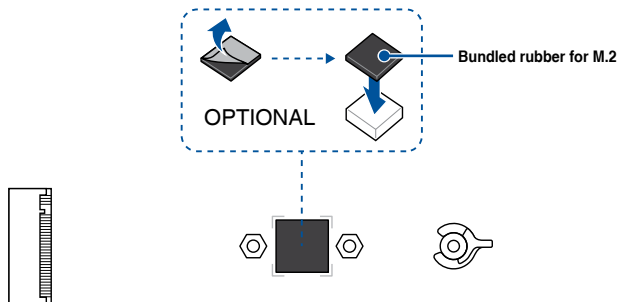
1. Loosen the screws from the M.2 heatsink.
2. Lift and remove the heatsinks.



3. (optional) Install the bundled rubber for M.2 if you are installing a single sided M.2 module. **DO NOT** install the bundled rubber for M.2 when installing a double-sided M.2 module. The rubber installed by default is compatible with double sided M.2 modules.



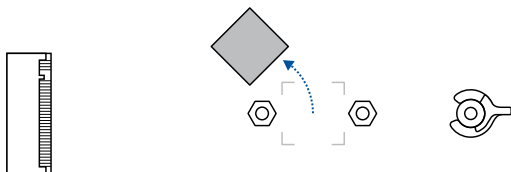
Only follow this step if installing a 2280 length M.2 module and when the rubber for M.2 comes bundled with your motherboard package.



OR remove the M.2 rubber.



Only follow this step if installing a 2242 length M.2 module and the M.2 slot has an M.2 rubber pre-installed.

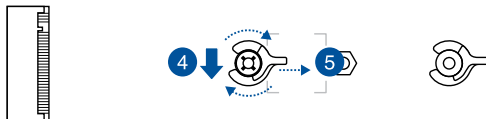


4. (optional) Install the M.2 Q-Latch to the M.2 length screw hole you wish to install your M.2 module to.



You can use a bundled M.2 Q-Latch screw or a pre-installed removable M.2 Q-Latch screw.

5. Rotate and adjust the M.2 Q-Latch so that the handle points away from the M.2 slot.

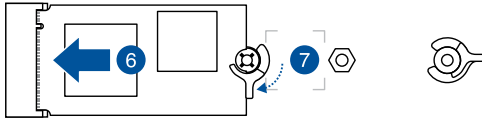


6. Install your M.2 module to the M.2 slot.



Ensure that there is nothing obstructing your M.2 module when installing the M.2 module to the M.2 slot.

7. Rotate the M.2 Q-Latch clockwise to secure the M.2 module in place.

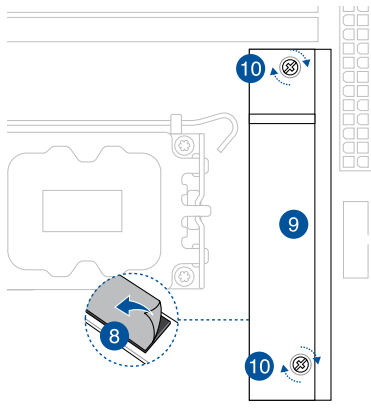


8. Remove the plastic film from the thermal pads on the bottom of the heatsinks.



If the thermal pad on the M.2 heatsink becomes damaged, we recommend replacing it with the bundled thermal pad or a thermal pad with a thickness of 1.25mm.

9. Replace the heatsink.
10. Secure the heatsink using the screws on the heatsink.

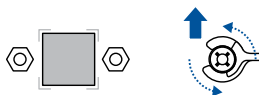


For M.2_2 (Type 2242 / 2260 / 2280 M.2)

1. (optional) If required, remove the pre-installed removable M.2 Q-Latch screw at the 2280 length screw hole.



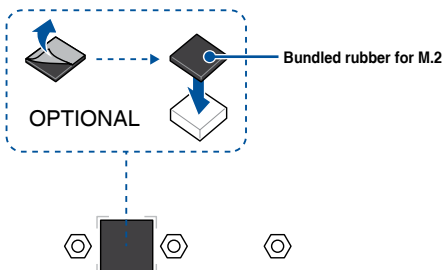
Only follow this step if a removable M.2 Q-Latch screw is pre-installed at the 2280 length screw hole and can be removed.



2. (optional) Install the bundled rubber for M.2 if you are installing a single sided M.2 module. **DO NOT** install the bundled rubber for M.2 when installing a double-sided M.2 module. The rubber installed by default is compatible with double sided M.2 modules.



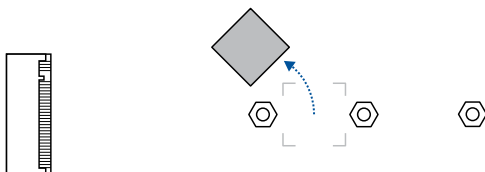
Only follow this step if installing a 22110 or 2280 length M.2 module and when the rubber for M.2 comes bundled with your motherboard package.



OR remove the M.2 rubber.



Only follow this step if installing a 2242 length M.2 module and the M.2 slot has an M.2 rubber pre-installed.



- (optional) Install the M.2 Q-Latch to the M.2 length screw hole you wish to install your M.2 module to.



You can use a bundled M.2 Q-Latch screw or a pre-installed removable M.2 Q-Latch screw.

- Rotate and adjust the M.2 Q-Latch so that the handle points away from the M.2 slot.

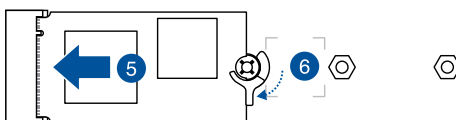


- Install your M.2 module to the M.2 slot.



Ensure that there is nothing obstructing your M.2 module when installing the M.2 module to the M.2 slot.

- Rotate the M.2 Q-Latch clockwise to secure the M.2 module in place.

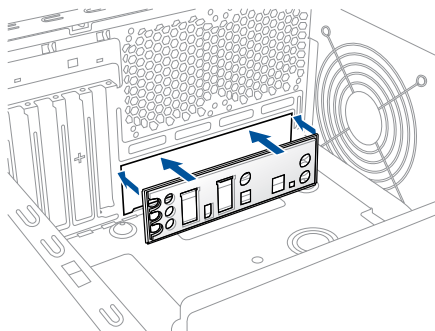


2.4 Motherboard installation

1. (on selected models) Install the bundled I/O Shield to the chassis rear I/O panel.



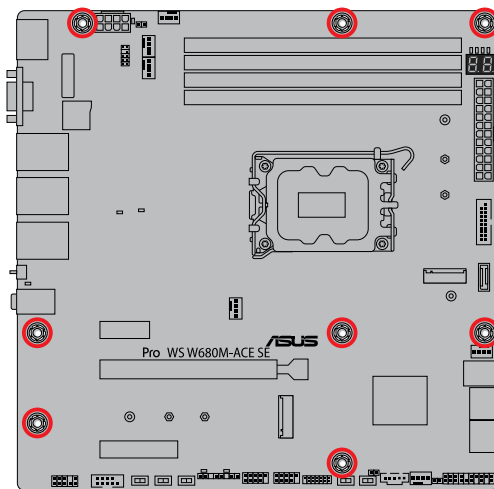
Only install the I/O Shield if your motherboard does not have a pre-installed I/O shield.



2. Place the motherboard into the chassis, ensuring that its rear I/O ports are aligned to the chassis' rear I/O panel.
3. Place eight (8) screws into the holes indicated by circles to secure the motherboard to the chassis.



This instruction is for reference only, please place the amount of screws according to your installation situation.



DO NOT over tighten the screws! Doing so can damage the motherboard.

2.5 BIOS FlashBack™



The illustrations for this section are for reference only. The WiFi module is only available on selected models.

BIOS FlashBack™ allows you to easily update the BIOS without entering the existing BIOS or operating system.

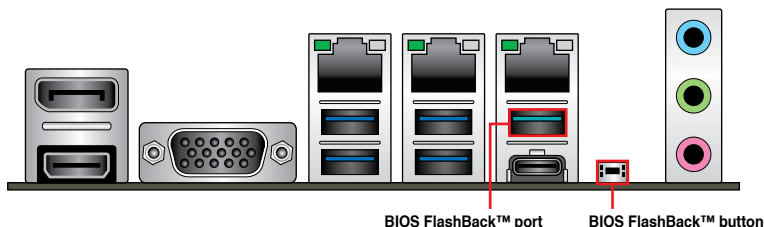
To use BIOS FlashBack™:

1. Visit <https://www.asus.com/support/> and download the latest BIOS version for this motherboard.
2. Launch the **BIOSRenamer.exe** application to automatically rename the file or manually rename the file to the BIOS CAP filename specified in the **Specifications summary** section, then copy it to your USB storage device.



The **BIOSRenamer.exe** application is zipped together with your BIOS file when you download a BIOS file for a BIOS FlashBack™ compatible motherboard.

3. Plug the 24-pin power connector to the motherboard and turn on the power supply (no need to power on the system). Insert the USB storage device to the USB port with BIOS FlashBack™ function.
4. Press the BIOS FlashBack™ button for three (3) seconds until the BIOS FlashBack™ LED blinks three times, indicating that the BIOS FlashBack™ function is enabled.



5. Wait until the light goes out, indicating that the BIOS updating process is completed.



For more BIOS update utilities in BIOS setup, refer to **BIOS and RAID Support** section.



- Do not unplug portable disk, power system, press the Clear CMOS button, or short the CLRRTC header while BIOS update is ongoing, otherwise update will be interrupted. In case of interruption, please follow the steps again.
- If the light flashes for five seconds and turns into a solid light, this means that the BIOS FlashBack™ is not operating properly. This may be caused by improper installation of the USB storage device and filename/file format error. If this scenario happens, please restart the system to turn off the light.
- Updating BIOS may have risks. If the BIOS program is damaged during the process and results to the system's failure to boot up, please contact your local ASUS Service Center.

2.6 Starting up for the first time

1. After making all the connections, replace the system case cover.
2. Ensure that all switches are off.
3. Connect the power cord to the power connector at the back of the system chassis.
4. Connect the power cord to a power outlet that is equipped with a surge protector.
5. Turn on the devices in the following order:
 - a. Monitor
 - b. External storage devices (starting with the last device on the chain)
 - c. System power
6. After applying power, the system power LED on the system front panel case lights up. For systems with ATX power supplies, the system LED lights up when you press the ATX power button. If your monitor complies with the “green” standards or if it has a “power standby” feature, the monitor LED may light up or change from orange to green after the system LED turns on.

The system then runs the power-on self tests (POST). While the tests are running, the BIOS beeps (refer to the BIOS beep codes table) or additional messages appear on the screen. If you do not see anything within 30 seconds from the time you turned on the power, the system may have failed a power-on test. Check the jumper settings and connections or call your retailer for assistance.

BIOS Beep	Description
One short beep	VGA detected Quick boot set to disabled No keyboard detected
One continuous beep followed by two short beeps then a pause (repeated)	No memory detected
One continuous beep followed by three short beeps	No VGA detected
One continuous beep followed by four short beeps	Hardware component failure

7. At power on, hold down the <Delete> key to enter the BIOS Setup. Follow the instructions in Chapter 3.

2.7 Turning off the computer

While the system is ON, press the power button for less than four seconds to put the system on sleep mode or soft-off mode, depending on the BIOS setting. Press the power button for more than four seconds to let the system enter the soft-off mode regardless of the BIOS setting.

[illegible]

BIOS and RAID Support

3



For more details on BIOS and RAID configurations, please refer to Manual & Document under the Support tab of the product information site, or visit <https://www.asus.com/support>.

3.1 Knowing UEFI BIOS

BIOS (Basic Input and Output System) stores system hardware settings such as storage device configuration, overclocking settings, advanced power management, and boot device configuration that are needed for system startup in the motherboard CMOS. In normal circumstances, the default BIOS settings apply to most conditions to ensure optimal performance. **DO NOT change the default BIOS settings** except in the following circumstances:

- An error message appears on the screen during the system bootup and requests you to run the BIOS Setup.
- You have installed a new system component that requires further BIOS settings or update.



Inappropriate BIOS settings may result in instability or boot failure. **We strongly recommend that you change the BIOS settings only with the help of a trained service personnel.**



BIOS settings and options may vary due to different BIOS release versions. Please refer to the latest BIOS version for settings and options.

Entering BIOS at startup

To enter BIOS Setup at startup, press <Delete> or <F2> during the Power-On Self Test (POST). If you do not press <Delete> or <F2>, POST continues with its routines.



- If the system becomes unstable after changing any BIOS setting, load the default settings to ensure system compatibility and stability. Select the **Load Optimized Defaults** item under the **Exit** menu or press the <F5> hotkey.
- If the system fails to boot after changing any BIOS setting, try to clear the CMOS and reset the motherboard to the default value.
- The BIOS setup program does not support Bluetooth devices.

3.2 ASUS EZ Flash Utility

The ASUS EZ Flash Utility feature allows you to update the BIOS without using an OS-based utility.



Ensure to load the BIOS default settings to ensure system compatibility and stability. Select the **Load Optimized Defaults** item under the **Exit** menu or press hotkey <F5>.

To update the BIOS:



- This function can support devices such as a USB flash disk with FAT 32/16 format and single partition only.
- DO NOT shut down or reset the system while updating the BIOS to prevent system boot failure!

1. Insert the USB flash disk that contains the latest BIOS file to the USB port.
2. Enter the Advanced Mode of the BIOS setup program. Go to the **Tool** menu to select **Start ASUS EzFlash** and press <Enter>.
3. Press the Left arrow key to switch to the **Drive** field.
4. Press the Up/Down arrow keys to find the USB flash disk that contains the latest BIOS, and then press <Enter>.
5. Press the Right arrow key to switch to the **Folder** field.
6. Press the Up/Down arrow keys to find the BIOS file, and then press <Enter> to perform the BIOS update process. Reboot the system when the update process is done.

3.3 ASUS CrashFree BIOS 3

The ASUS CrashFree BIOS 3 utility is an auto recovery tool that allows you to restore the BIOS file when it fails or gets corrupted during the updating process. You can restore a corrupted BIOS file using a USB flash drive that contains the BIOS file.

Recovering the BIOS

1. Download the latest BIOS version for this motherboard from <https://www.asus.com/support/>.
2. Rename the file using one of the following methods:
 - Launch the **BIOSRenamer.exe** application to automatically rename the file.
 - Manually rename the file to the BIOS CAP filename specified in the **Specifications summary** section.
 - Manually rename the file to **asus.cap**.
3. Copy the renamed file to your USB storage device.
4. Turn on the system.
5. Insert the USB flash drive containing the BIOS file to a USB port.
6. The utility automatically checks the devices for the BIOS file. When found, the utility reads the BIOS file and enters ASUS EZ Flash Utility automatically.
7. The system requires you to enter BIOS Setup to recover the BIOS setting. To ensure system compatibility and stability, we recommend that you press <F5> to load default BIOS values.



DO NOT shut down or reset the system while updating the BIOS! Doing so can cause system boot failure!

3.4 RAID configurations

The motherboard supports RAID configurations.

RAID definitions

RAID 0 (Data striping) optimizes two identical hard disk drives to read and write data in parallel, interleaved stacks. Two hard disks perform the same work as a single drive but at a sustained data transfer rate, double that of a single disk alone, thus improving data access and storage. Use of two new identical hard disk drives is required for this setup.

RAID 1 (Data mirroring) copies and maintains an identical image of data from one drive to a second drive. If one drive fails, the disk array management software directs all applications to the surviving drive as it contains a complete copy of the data in the other drive. This RAID configuration provides data protection and increases fault tolerance to the entire system. Use two new drives or use an existing drive and a new drive for this setup. The new drive must be of the same size or larger than the existing drive.

RAID 5 stripes both data and parity information across three or more hard disk drives. Among the advantages of RAID 5 configuration include better HDD performance, fault tolerance, and higher storage capacity. The RAID 5 configuration is best suited for transaction processing, relational database applications, enterprise resource planning, and other business systems. Use a minimum of three identical hard disk drives for this setup.

RAID 10 is data striping and data mirroring combined without parity (redundancy data) having to be calculated and written. With the RAID 10 configuration you get all the benefits of both RAID 0 and RAID 1 configurations. Use four new hard disk drives or use an existing drive and three new drives for this setup.

Appendix

Q-Code table

Code	Description
00	Not used
01	Power on. Reset type detection (soft/hard).
02	AP initialization before microcode loading
03	System Agent initialization before microcode loading
04	PCH initialization before microcode loading
06	Microcode loading
07	AP initialization after microcode loading
08	System Agent initialization after microcode loading
09	PCH initialization after microcode loading
0B	Cache initialization
0C – 0D	Reserved for future AMI SEC error codes
0E	Microcode not found
0F	Microcode not loaded
10	PEI Core is started
11 – 14	Pre-memory CPU initialization is started
15 – 18	Pre-memory System Agent initialization is started
19 – 1C	Pre-memory PCH initialization is started
2B – 2F	Memory initialization
30	Reserved for ASL (see ASL Status Codes section below)
31	Memory Installed
32 – 36	CPU post-memory initialization
37 – 3A	Post-Memory System Agent initialization is started
3B – 3E	Post-Memory PCH initialization is started
4F	DXE IPL is started
50 – 53	Memory initialization error. Invalid memory type or incompatible memory speed
54	Unspecified memory initialization error
55	Memory not installed
56	Invalid CPU type or Speed
57	CPU mismatch
58	CPU self test failed or possible CPU cache error
59	CPU micro-code is not found or micro-code update is failed
5A	Internal CPU error
5B	Reset PPI is not available
5C – 5F	Reserved for future AMI error codes

(continued on the next page)

Q-Code table

Code	Description
E0	S3 Resume is started (S3 Resume PPI is called by the DXE IPL)
E1	S3 Boot Script execution
E2	Video repost
E3	OS S3 wake vector call
E4 – E7	Reserved for future AML progress codes
E8	S3 Resume Failed
E9	S3 Resume PPI not Found
EA	S3 Resume Boot Script Error
EB	S3 OS Wake Error
EC – EF	Reserved for future AML error codes
F0	Recovery condition triggered by firmware (Auto recovery)
F1	Recovery condition triggered by user (Forced recovery)
F2	Recovery process started
F3	Recovery firmware image is found
F4	Recovery firmware image is loaded
F5 – F7	Reserved for future AML progress codes
F8	Recovery PPI is not available
F9	Recovery capsule is not found
FA	Invalid recovery capsule
FB – FF	Reserved for future AML error codes
60	DXE Core is started
61	NVRAM initialization
62	Installation of the PCH Runtime Services
63 – 67	CPU DXE initialization is started
68	PCI host bridge initialization
69	System Agent DXE initialization is started
6A	System Agent DXE SMM initialization is started
6B – 6F	System Agent DXE initialization (System Agent module specific)
70	PCH DXE initialization is started
71	PCH DXE SMM initialization is started
72	PCH devices initialization
73 – 77	PCH DXE Initialization (PCH module specific)
78	ACPI module initialization
79	CSM initialization
7A – 7F	Reserved for future AML DXE codes

(continued on the next page)

Q-Code table

Code	Description
90	Boot Device Selection (BDS) phase is started
91	Driver connecting is started
92	PCI Bus initialization is started
93	PCI Bus Hot Plug Controller Initialization
94	PCI Bus Enumeration
95	PCI Bus Request Resources
96	PCI Bus Assign Resources
97	Console Output devices connect
98	Console input devices connect
99	Super IO Initialization
9A	USB initialization is started
9B	USB Reset
9C	USB Detect
9D	USB Enable
9E – 9F	Reserved for future AMI codes
A0	IDE initialization is started
A1	IDE Reset
A2	IDE Detect
A3	IDE Enable
A4	SCSI initialization is started
A5	SCSI Reset
A6	SCSI Detect
A7	SCSI Enable
A8	Setup Verifying Password
A9	Start of Setup
AA	Reserved for ASL (see ASL Status Codes section below)
AB	Setup Input Wait
AC	Reserved for ASL (see ASL Status Codes section below)
AD	Ready To Boot event
AE	Legacy Boot event
AF	Exit Boot Services event
B0	Runtime Set Virtual Address MAP Begin
B1	Runtime Set Virtual Address MAP End
B2	Legacy Option ROM Initialization
B3	System Reset

(continued on the next page)

Q-Code table

Code	Description
B4	USB hot plug
B5	PCI bus hot plug
B6	Clean-up of NVRAM
B7	Configuration Reset (reset of NVRAM settings)
B8– BF	Reserved for future AML codes
D0	CPU initialization error
D1	System Agent initialization error
D2	PCH initialization error
D3	Some of the Architectural Protocols are not available
D4	PCI resource allocation error. Out of Resources
D5	No Space for Legacy Option ROM
D6	No Console Output Devices are found
D7	No Console Input Devices are found
D8	Invalid password
D9	Error loading Boot Option (LoadImage returned error)
DA	Boot Option is failed (StartImage returned error)
DB	Flash update is failed
DC	Reset protocol is not available

ACPI/ASL Checkpoints (under OS)

Code	Description
03	System is entering S3 sleep state
04	System is entering S4 sleep state
05	System is entering S5 sleep state
30	System is waking up from the S3 sleep state
40	System is waking up from the S4 sleep state
AC	System has transitioned into ACPI mode. Interrupt controller is in PIC mode.
AA	System has transitioned into ACPI mode. Interrupt controller is in APIC mode.

General Notices

FCC Compliance Information

Responsible Party: Asus Computer International

Address: 48720 Kato Rd., Fremont, CA 94538, USA

Phone / Fax No: (510)739-3777 / (510)608-4555

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

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HIGH-DEFINITION MULTIMEDIA INTERFACE

Safety Precautions

Accessories that came with this product have been designed and verified for the use in connection with this product. Never use accessories for other products to prevent the risk of electric shock or fire.

安全上のご注意

付属品は当該専用品です。他の機器には使用しないでください。機器の破損もしくは、火災や感電の原因となることがあります。

VCCI: Japan Compliance Statement

Class B ITE

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取扱説明書に従って正しい取り扱いをして下さい。

VCCI-B

Japan JATE

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Declaration of compliance for product environmental regulation

ASUS follows the green design concept to design and manufacture our products, and makes sure that each stage of the product life cycle of ASUS product is in line with global environmental regulations. In addition, ASUS disclose the relevant information based on regulation requirements.

Please refer to <https://csr.asus.com/Compliance.htm> for information disclosure based on regulation requirements ASUS is complied with:

EU REACH and Article 33

Complying with the REACH (Registration, Evaluation, Authorisation, and Restriction of Chemicals) regulatory framework, we published the chemical substances in our products at ASUS REACH website at <https://csr.asus.com/english/REACH.htm>.

EU RoHS

This product complies with the EU RoHS Directive. For more details, see <https://csr.asus.com/english/article.aspx?id=35>

India RoHS

This product complies with the "India E-Waste (Management) Rules, 2016" and prohibits use of lead, mercury, hexavalent chromium, polybrominated biphenyls (PBBs) and polybrominated diphenyl ethers (PBDEs) in concentrations exceeding 0.1% by weight in homogenous materials and 0.01% by weight in homogenous materials for cadmium, except for the exemptions listed in Schedule II of the Rule.

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ASUS products sold in Vietnam, on or after September 23, 2011, meet the requirements of the Vietnam Circular 30/2011/TT-BCT.

Các sản phẩm ASUS bán tại Việt Nam, vào ngày 23 tháng 9 năm 2011 trở về sau, đều phải đáp ứng các yêu cầu của Thông tư 30/2011/TT-BCT của Việt Nam.

Türkiye RoHS

AEEE Yönetmeliğine Uygundur

ASUS Recycling/Takeback Services

ASUS recycling and takeback programs come from our commitment to the highest standards for protecting our environment. We believe in providing solutions for you to be able to responsibly recycle our products, batteries, other components as well as the packaging materials. Please go to <https://csr.asus.com/english/Takeback.htm> for detailed recycling information in different regions.



DO NOT throw the motherboard in municipal waste. This product has been designed to enable proper reuse of parts and recycling. This symbol of the crossed out wheeled bin indicates that the product (electrical and electronic equipment) should not be placed in municipal waste. Check local regulations for disposal of electronic products.



DO NOT throw the mercury-containing button cell battery in municipal waste. This symbol of the crossed out wheeled bin indicates that the battery should not be placed in municipal waste.

France sorting and recycling information



Points de collecte sur www.quefairedemesdechets.fr
Privilégiez la réparation ou le don de votre appareil !

Notices for non Wi-Fi model

Compliance Statement of Innovation, Science and Economic Development Canada (ISED)

This device complies with Innovation, Science and Economic Development Canada licence exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

CAN ICES-003(B)/NMB-003(B)

Déclaration de conformité de Innovation, Sciences et Développement économique Canada (ISED)

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CAN ICES-003(B)/NMB-003(B)

KC: Korea Warning Statement

B급 기기 (가정용 방송통신기자재)

이 기기는 가정용(B급) 전자파적합기기로서 주로 가정에서 사용하는 것을 목적으로 하며, 모든 지역에서 사용할 수 있습니다.

Simplified UKCA Declaration of Conformity

ASUSTeK Computer Inc. hereby declares that this device is in compliance with the essential requirements and other relevant provisions of related UKCA Directives. Full text of UKCA declaration of conformity is available at: www.asus.com/support

Simplified EU Declaration of Conformity

English ASUSTeK Computer Inc. hereby declares that this device is in compliance with the essential requirements and other relevant provisions of related Directives. Full text of EU declaration of conformity is available at: www.asus.com/support

Français ASUSTeK Computer Inc. déclare par la présente que cet appareil est conforme aux critères essentiels et autres clauses pertinentes des directives concernées. La déclaration de conformité de l'UE peut être téléchargée à partir du site Internet suivant : www.asus.com/support

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- ASUS offers a voluntary manufacturer's Commercial Guarantee.
- ASUS reserves the right to interpret the provisions of the ASUS Commercial Guarantee.
- This ASUS Commercial Guarantee is provided independently and in addition to the statutory Legal Guarantee and in no way affects or limits the rights under the Legal Guarantee.

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- ASUS offre una Garanzia Commerciale volontaria del produttore.
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- ASUS forbeholder sig retten til at fortolke bestemmelse i ASUS' handelsmæssige garanti.
- Denne handelsmæssige garanti fra ASUS tilbydes uafhængigt, som en tilføjelse til den lovbestemte juridiske garanti og den påvirker eller begrænser på ingen måde rettighederne i den juridiske garanti.

Alle garantioplysningerne kan findes på <https://www.asus.com/dk/support/>.

BG: Информация за гаранцията от ASUS

- ASUS предлага доброволна търговска гаранция от производителя.
- ASUS си запазва правото да тълкува условията на търговската гаранция на ASUS.
- Тази търговска гаранция на ASUS се предлага независимо от и в допълнение на законовата гаранция. Тя по никакъв начин не оказва влияние върху правата на потребителя в законовата гаранция и по никакъв начин не ги ограничава.

За цялостна информация относно гаранцията, моля, посетете <https://www.asus.com/bg/support/>.

CZ: Informace o záruce společnosti ASUS

- Společnost ASUS nabízí dobrovolnou komerční záruku výrobce.
- Společnost ASUS si vyhradzuje právo vykládat ustanovení komerční záruky společnosti ASUS.
- Tato komerční záruka společnosti ASUS je poskytována nezávisle a jako doplněk zákonné záruky a žádným způsobem neovlivňuje ani neomezuje práva vyplývající ze zákonné záruky.

Všechny informace o záruce najdete na adrese <https://www.asus.com/cz/support/>.

CR: Informacije o ASUS jamstvu

- ASUS dragovaljno nudi komercijalno proizvođačko jamstvo.
- ASUS zadržava prava na tumačenje odredbi ASUS komercijalnog jamstva.
- Ovo ASUS komercijalno jamstvo daje se neovisno i kao dodatak zakonskom jamstvu i ni na koji način ne ograničuje prava iz okvira zakonskog jamstva.

Sve informacije o jamstvu potražite na <https://www.asus.com/support/>.

DU: ASUS-garantie-informatie

- SUS biedt een vrijwillige commerciële garantie van de fabrikant.
- ASUS behoudt zich het recht voor om de bepalingen van de commerciële garantie van ASUS uit te leggen.
- Deze commerciële garantie van ASUS wordt onafhankelijk en als aanvulling op de statutaire Wettelijke garantie geboden en beïnvloedt of beperkt in geen geval de rechten onder de wettelijke garantie.

Voor alle informatie over de garantie, gaat u naar <https://www.asus.com/nl/support/>.

EE: Teave ASUS-e garantii kohta

- ASUS pakub vabatahtlikku tasulist tootjagarantiid.
- ASUS jätab endale õiguse tõlgendada ASUS-e tasulise garantii tingimusi.
- See ASUS-e tasuline garantiid on sõltumatu lisagarantiid seadusega kehtestatud garantiile ega mõjuta mingil määral seadusega kehtestatud garantiid ning seadusega kehtestatud garantiid piiranguid.

Vaadake garantiita seotud teavet veebisaidil <https://www.asus.com/ee/>.

GK: Πληροφορίες εγγύησης ASUS

- Η ASUS προσφέρει μια εθελοντική Εμπορική εγγύηση κατασκευαστή.
- Η ASUS διατηρεί το δικαίωμα ερμηνείας των διατάξεων της Εμπορικής εγγύησης ASUS.
- Αυτή η Εμπορική εγγύηση ASUS παρέχεται ανεξάρτητα και επιπροσθέτως της θεσμικής Νομικής εγγύησης και σε καμία περίπτωση δεν επηρεάζει ή περιορίζει τα δικαιώματα βάσει της Νομικής εγγύησης.

Για όλες τις πληροφορίες εγγύησης, επισκεφθείτε τη διεύθυνση <https://www.asus.com/gr/el/>.

HUG: ASUS garanciāls informācija

- Az ASUS önkéntes gyártói kereskedelmi garanciát kínál.
- Az ASUS fenntartja magának a jogot, hogy értelmezze az ASUS kereskedelmi garanciára vonatkozó rendelkezéseket.
- Ezt a kereskedelmi garanciát az ASUS függetlenül és a törvényes garancia mellett nyújtja és semmilyen módon nem befolyásolja, vagy korlátozza a jogi garancia nyújtotta jogokat.

A garanciára vonatkozó teljes körű információkért látogasson el a <https://www.asus.com/hu/support/oldala/>.

LV: ASUS garantijas informācija

- ASUS piedāvā brīvprātīgu ražotāja komerciālo garantiju.
- ASUS patur tiesības interpretēt ASUS komerciālās garantijas noteikumus.
- Šī ASUS komerciālā garantija tiek piedāvāta neatkarīgi un papildus likumā noteiktajai juridiskajai garantijai, un tā nekādā neietekmē vai neierobežo juridiskajā garantijā noteiktās tiesības.

Lai iegūtu informāciju par garantiju, apmeklējiet vietni <https://www.asus.com/lv/>.

LT: Informacija apie ASUS garantiją

- ASUS siūlo savanorišką komercinę gamintojo garantiją.
- ASUS pasilieka teisę savo nuostatais aiškinti šios komercinės ASUS garantijos nuostatas.
- Ši komercinė ASUS garantija suteikiama nepriklausoma, be įstatymine teisine garantijos, ir jokiu būdu nepaveikia ar neapriboja teisine garantijos suteikiamų teisių.

Norėdami gauti visą informaciją apie garantiją, apsilankykite <https://www.asus.com/lt/>.

PL: Informacje o gwarancji firmy ASUS

- Firma ASUS oferuje dobrowolną gwarancję handlową producenta.
- Firma ASUS zastrzega sobie prawo do interpretacji warunków gwarancji handlowej firmy ASUS.
- Niniejsza gwarancja handlowa firmy ASUS jest udzielana niezależnie, jako dodatek do wymaganej ustawowo gwarancji prawnej i w żaden sposób nie wpływa na prawa przysługujące na mocy gwarancji prawnej ani ich nie ogranicza.

Wszelkie informacje na temat gwarancji można znaleźć na stronie <https://www.asus.com/pl/support/>.

PG: Informações de Garantia ASUS

- A ASUS oferece uma Garantia Comercial voluntária do fabricante.
- A ASUS reserva o direito de interpretar as disposições da Garantia Comercial da ASUS.
- Esta Garantia Comercial da ASUS é fornecida de forma independente além da Garantia Legal estatutária e não afeta nem limita de qualquer forma os direitos estabelecidos na Garantia Legal.

Para consultar todas as informações sobre a garantia, visite <https://www.asus.com/pt/support/>.

RO: Informații despre garanția ASUS

- ASUS oferă o garanție comercială voluntară a producătorului.
- ASUS își rezervă dreptul de a interpreta prevederile garanției comerciale ASUS.
- Această garanție comercială ASUS este oferită independent și în plus față de garanția obligatorie legală și nu afectează sau limitează în niciun fel drepturile acordate conform garanției legale.

Pentru toate informațiile legate de garanție, vizitați <https://www.asus.com/ro/support>.

SL: Informacije o garanciji ASUS

- ASUS ponuja prostovoljno tržno garancijo proizvajalca.
- ASUS si pridržuje pravico do razlage določb tržne garancije družbe ASUS.
- Ta tržna garancija družbe ASUS je na voljo neodvisno in kot dodatek zakonsko predpisani pravni garanciji ter na noben način ne vpliva na pravice, ki jih zagotavlja pravna garancija, oziroma jih omejuje.

Vse informacije o garanciji najdete na spletnem mestu <https://www.asus.com/support>.

SK: Informácie o záruke ASUS

- ASUS ponúka dobrovoľnú obchodnú záruku výrobcu.
- ASUS si vyhradzuje právo interpretovať ustanovenia obchodnej záruky ASUS.
- Táto obchodná záruka ASUS je poskytnutá nezávisle a navyše k zákonnej záruke a v žiadnom prípade neovplyvňuje ani neobmedzuje tieto práva podľa tejto zákonnej záruky.

Všetky další informace o záruce najdete na <https://www.asus.com/sk/support>.

ES: Información de garantía de ASUS

- ASUS ofrece una garantía comercial voluntaria del fabricante.
- ASUS se reserva el derecho de interpretar las disposiciones de esta garantía comercial de ASUS.
- Esta garantía comercial de ASUS se proporciona de forma independiente y adicional a la garantía estatutaria y de ninguna manera afecta a los derechos bajo la garantía legal ni los limita.

Para obtener toda la información sobre la garantía, visite <https://www.asus.com/ES/support/>.

TR: ASUS Garanti Bilgileri

- ASUS, gönüllü olarak üretici Ticari Garantisini sunar.
- ASUS, ASUS Ticari Garantisinin hükümlerini yorumlama hakkını saklı tutar.
- Bu ASUS Ticari Garantisini, bağımsız olarak ve hukuki Yasal Garanti'ye ek olarak sağlanır ve hiçbir şekilde Yasal Garanti kapsamındaki hakları etkilemez veya sınırlanmaz.

Tüm garanti bilgileri için lütfen <https://www.asus.com/tr/support> adresini ziyaret edin.

FI: ASUS-takuutiedot

- ASUS tarjoaa vapaaehtoisien valmistajan kauppallisen takuun.
- ASUS pidättää oikeuden tulkita ASUS-kauppallisen takuun ehdot.
- Tämä ASUS-kauppallinen takuu tarjotaan itsenäisesti lakisääteisen oikeudellisen takuun lisäksi eikä se vaikuta millään tavoin laillisen takuun oikeuksiin tai rajoita niitä.

Saadaksesi kaikki takuutiedot, siirry osoitteeseen <https://www.asus.com/fin/support>.

NW: Informasjon om ASUS-garanti

- ASUS tilbyr som produsent en frivillig kommersiell garanti.
- ASUS forbeholder seg retten til å tolke bestemmelsene i ASUS sin kommersielle garanti.
- ASUS sin kommersielle garanti gir uavhengig og i tillegg til den lovbestemte juridiske garantien, og virken påvirker eller begrenser rettighetene under den juridiske garantien på noen måte.

Du finner fullstendig informasjon om garanti på <https://www.asus.com/no/support/>.

SB: Informacije o ASUS garanciji

- ASUS nudi dobrovoljno proizvođačku komercijalnu garanciju.
- ASUS zadržava pravo da tumači odredbe svoje ASUS komercijalne garancije.
- Ova ASUS komercijalna garancija daje se nezavisno, kao dodatak zakonskoj pravnoj garanciji, i ni na koji način ne utiče na i ne ograničava prava data pravnom garancijom.

Za sve informacije o garanciji, posetite <https://www.asus.com/support/>.

SW: ASUS garantiinformation

- ASUS erbjuder en frivillig kommersiell tillverkningsgaranti.
- ASUS förbehåller sig rätten att tolka bestämmelserna i ASUS kommersiella garanti.
- Denna kommersiella garanti från ASUS tillhandahålls separat och som tillägg till den lagstadgade garantin, och påverkar eller begränsar på intet sätt rättigheterna under den lagstadgade garantin.

För all garantiinformation, besök <https://www.asus.com/se/support/>.

UA: Інформація про Гарантію ASUS

- ASUS пропонує добровільну Комерційну Гарантію виробника.
- ASUS застерігає за собою право тлумачити положення Комерційної Гарантії ASUS
- Цю Комерційну Гарантію надано незалежно і на додаток до обов'язкової Законної Гарантії; вона жодним чином не впливає на права за Законною Гарантією і не обмежує їх.

Всю інформацію про гарантію подано тут:

<https://www.asus.com/ua/support>.

MX: Garantía y Soporte

Esta Garantía aplica en el país de compra. Usted acepta que en esta garantía:

- Los procedimientos de servicio pueden variar en función del país.
- Algunos servicios y/o piezas de reemplazo pudieran no estar disponibles en todos los países.
- Algunos países pueden tener tarifas y restricciones que se apliquen en el momento de realizar el servicio, visite el sitio de soporte de ASUS en <https://www.asus.com/mx/support/> para ver más detalles.
- Si tiene alguna queja o necesidad de un centro de reparación local o el periodo de garantía del producto ASUS, por favor visite el sitio de Soporte de ASUS en <https://www.asus.com/mx/support/> para mayores detalles.

Información de contacto ASUS

Esta garantía está respaldada por:
ASUSTeK Computer Inc.
Centro de Atención ASUS +52 (55) 1946-3663

BP: Informations de garantie ASUS

Esta garantía aplica-se ao período definido pela garantia legal (90 dias) mais o período de garantia comercial oferecido pela ASUS. Por exemplo: 12M significa 12 meses de garantia no total (3 meses de garantia legal mais 9 meses de garantia contratual), 24 meses significa 24 meses de garantia no total (3 meses de garantia legal mais 21 meses de garantia contratual) e 36 meses significa 36 meses de garantia no total (3 meses de garantia legal e 33 de garantia contratual) a contar da data da garantia declarada (Data de Início da Garantia).

Para todas as informações de garantia, visite <https://www.asus.com/br/support/>.

ID: Informasi Garansi ASUS

Garansi ini berlaku di negara tempat pembelian.

Periode Garansi tertera pada kemasan/kotak dari Produk dan Masa Garansi dimulai sejak tanggal pembelian Produk ASUS dengan kondisi baru.

Silahkan pindai Kode di bagian bawah halaman terakhir untuk Kartu Garansi versi Web dalam format PDF untuk lebih informasi jelas mengenai jaminan produk ASUS.

- Informasi Dukungan ASUS, silakan kunjungi <https://www.asus.com/id/support>.
- Informasi Lokasi Layanan, silakan kunjungi <https://www.asus.com/id/support/Service-Center/Indonesia>.
- Layanan Call Center: 1500128

VI: Thông tin đảm bảo của ASUS

- ASUS cung cấp Bảo hành thương mại tự nguyện của nhà sản xuất.
- ASUS bảo lưu quyền giải thích các điều khoản của Bảo hành thương mại của ASUS.
- Bảo hành thương mại này của ASUS được cung cấp độc lập và ngoài Bảo đảm pháp lý theo luật định và không có cách nào ảnh hưởng đến hoặc giới hạn các quyền theo Bảo lãnh pháp lý. Để biết tất cả các thông tin bảo hành, vui lòng truy cập

<https://www.asus.com/vn/support>



ASUS contact information

ASUSTeK COMPUTER INC.

Address: 1F., No. 15, Lide Rd., Beitou Dist., Taipei City 112

ASUS COMPUTER INTERNATIONAL (America)

Address: 48720 Kato Rd., Fremont, CA 94538, USA

ASUS COMPUTER GmbH (Germany and Austria)

Address: Harkortstrasse 21-23, 40880 Ratingen, Germany

ASUSTeK (UK) LIMITED

Address: 1st Floor, Sackville House, 143-149 Fenchurch Street, London, EC3M 6BL,
England, United Kingdom

Service and Support

Visit our multi-language website at <https://www.asus.com/support>.

